Predicting Urban Growth to Support Better Planning - Abuja, Nigeria

Quantity-constrained CA-Markov scenario to 2035

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Project overview

This project explores where Abuja may experience further urban expansion by 2035 and how that information can support earlier planning decisions. Historical land-cover transitions were combined with an urban-suitability surface and a quantity-constrained CA-Markov allocation model.

Planning question

Where is Abuja most likely to expand next, and how can planners prepare for that growth?

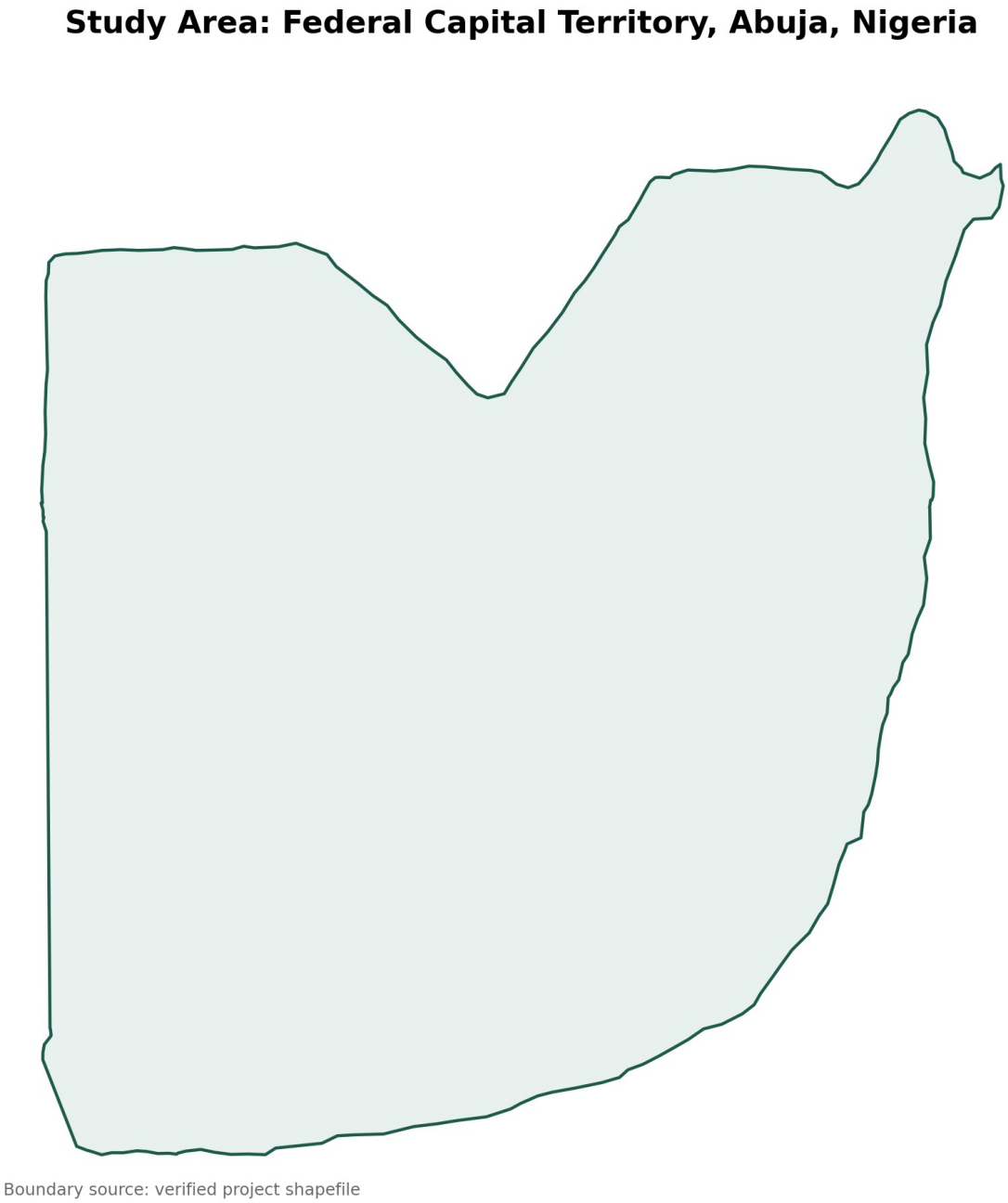


Figure 1. Federal Capital Territory study area.

Method

The workflow estimated land-cover transition probabilities from 2015 to 2025, calculated the quantity of built-up growth required for 2035, used an urban-suitability surface to rank candidate development locations, and allocated new built-up cells until the modelled demand was approached. The scenario rasters use a 30 m grid in WGS 84 / UTM Zone 32N (EPSG:32632).

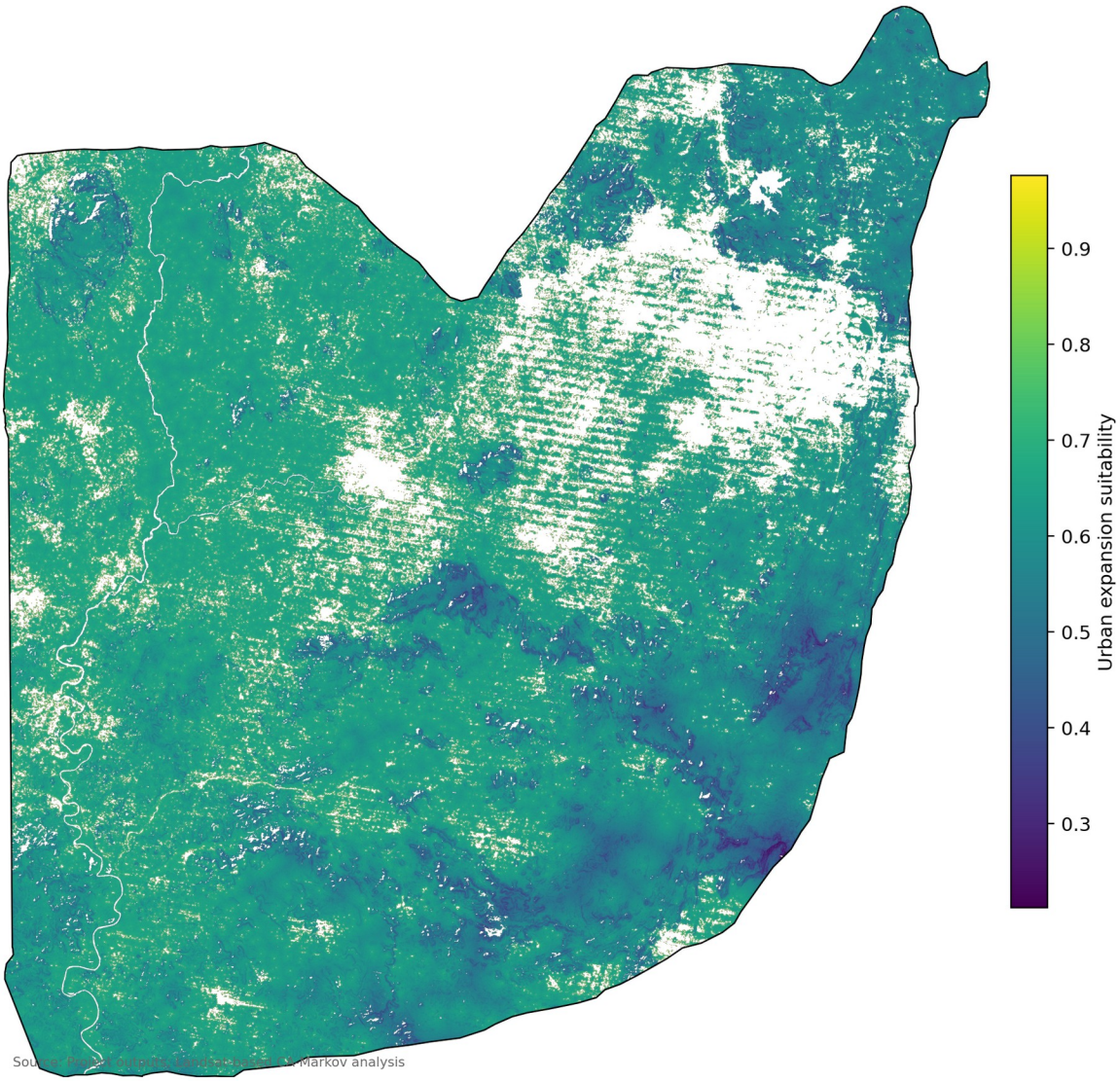


Figure 2. Urban-expansion suitability surface used for spatial allocation.

Scenario results

Indicator	Result
Built-up area, 2025	1,167.85 km²
Built-up area, 2035 scenario	1,351.30 km²
Net built-up increase	183.45 km²
Relative built-up increase	15.71%
Required new built-up pixels	206,027
Allocated new built-up pixels	203,889
Allocation difference	2,138 pixels

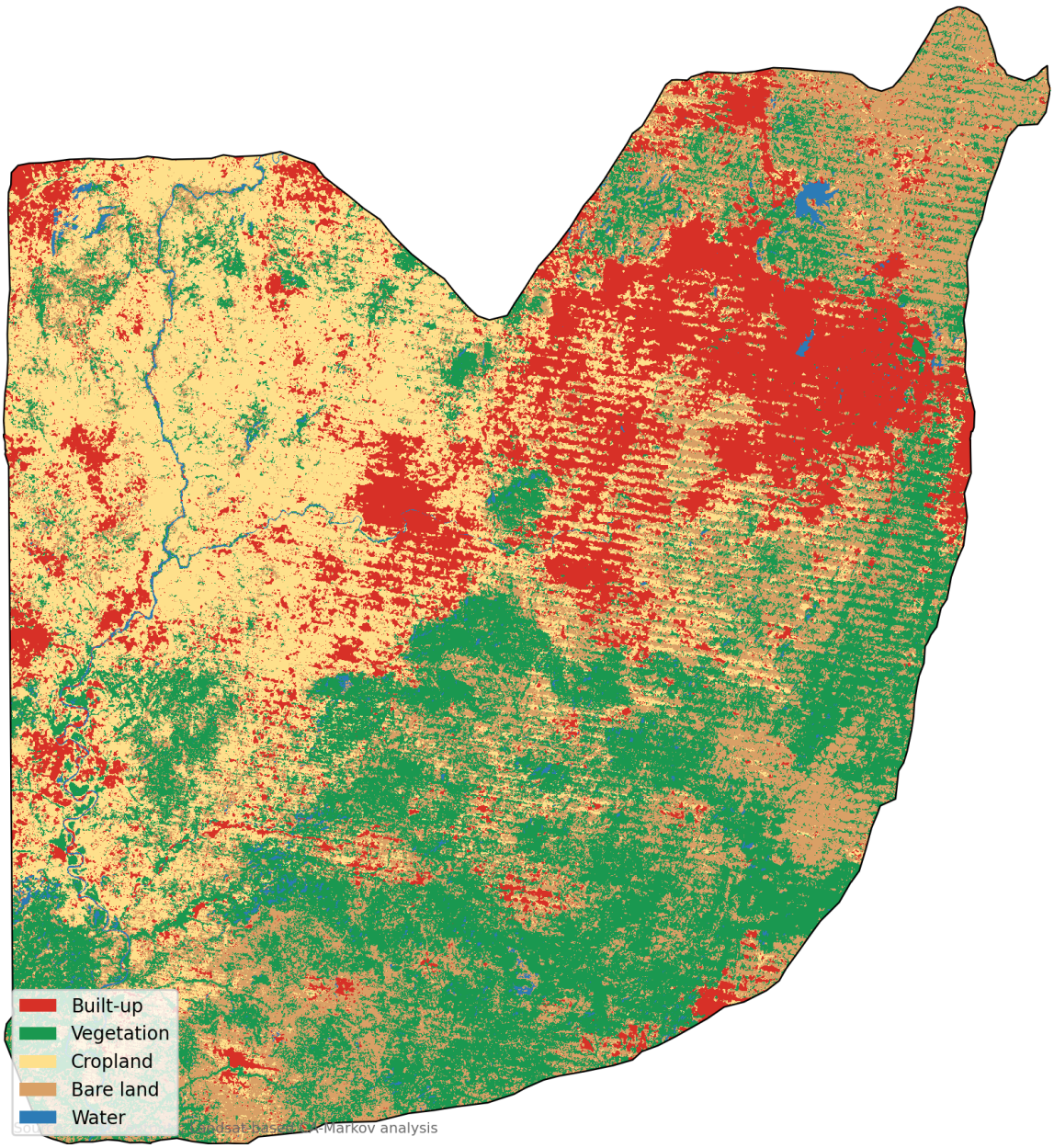


Figure 3. Modelled 2035 land-cover scenario.

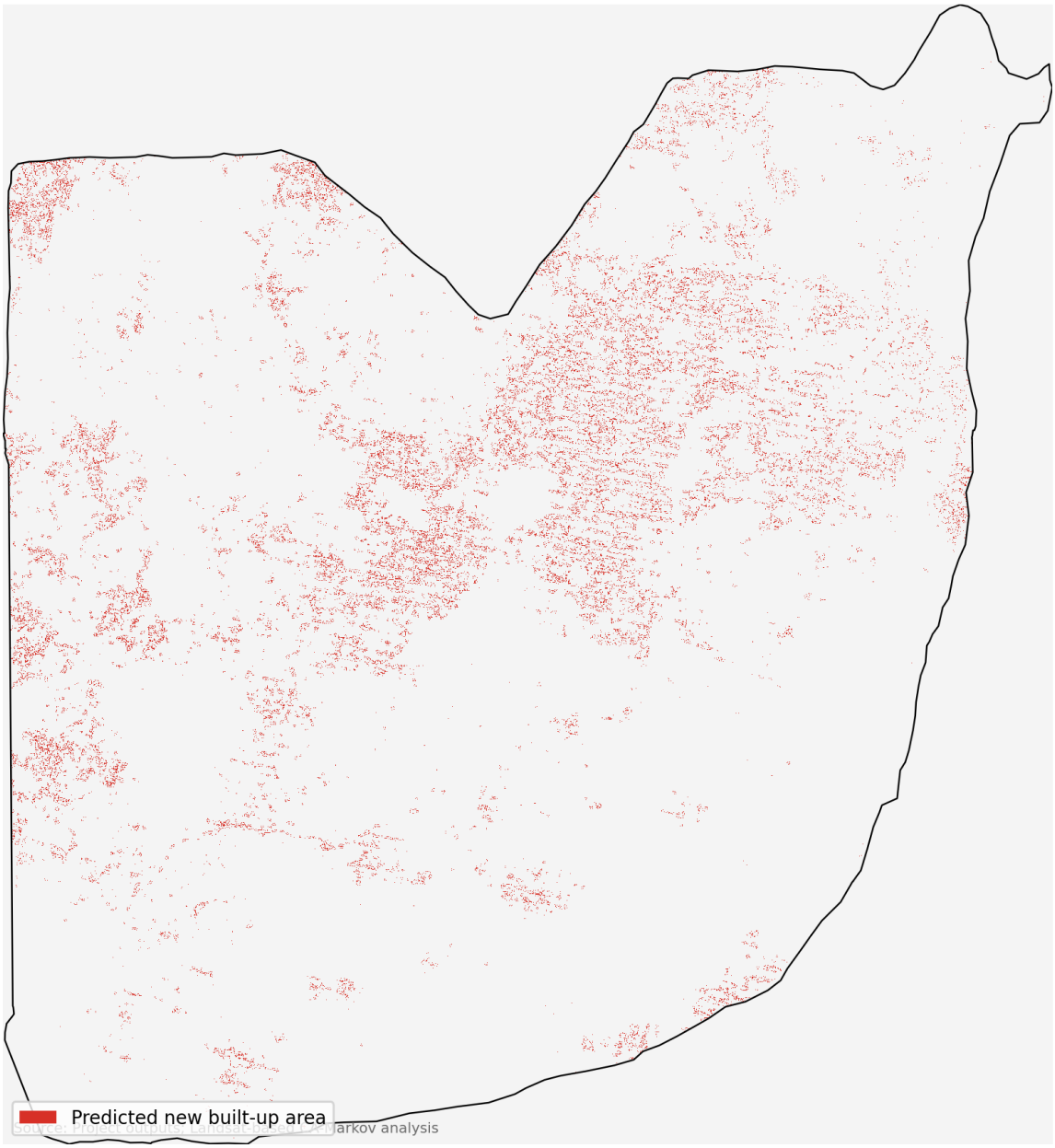


Figure 4. Modelled new built-up areas between 2025 and 2035.

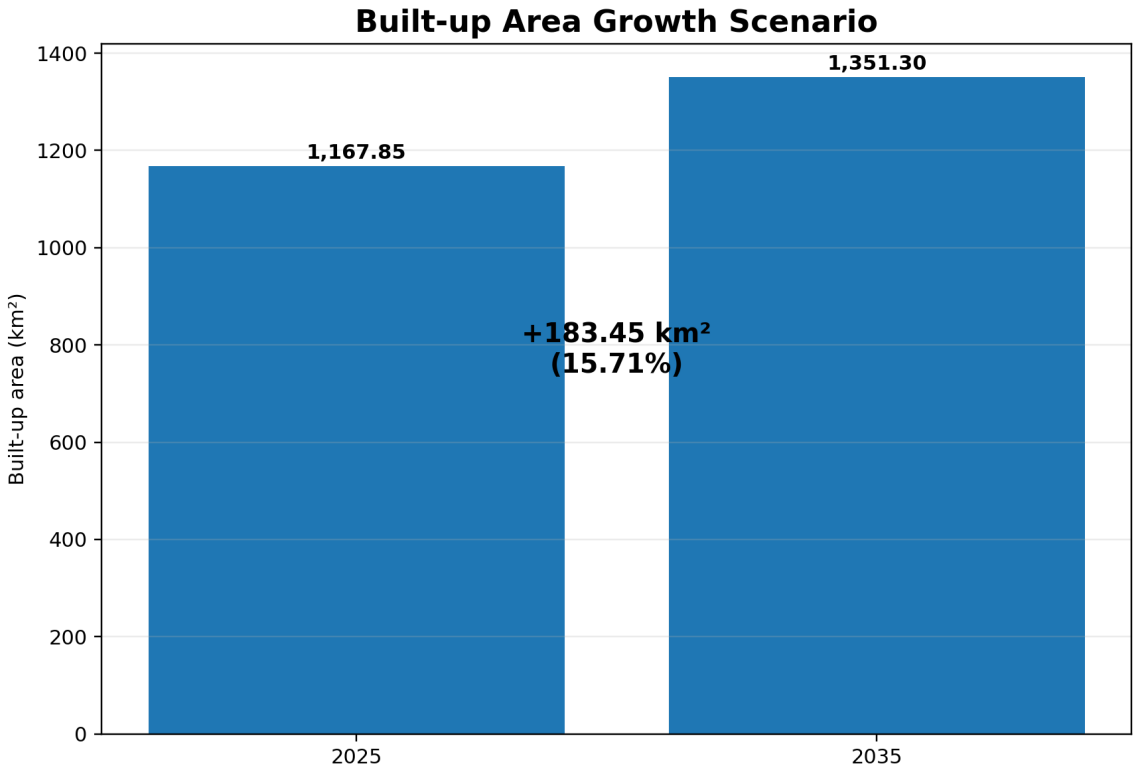


Figure 5. Built-up growth summary.

Validation and interpretation

Indicator	Result
Validation sample count	1,500
Overall Accuracy	0.4927 (49.27%)
Cohen's Kappa	0.3658

The historical validation indicates substantial model uncertainty. For that reason, the 2035 output should be treated as a strategic planning scenario rather than a certain forecast. Its strongest use is broad screening: identifying areas where infrastructure, development control and environmental safeguards may deserve earlier attention.

Planning value

- Use the expansion pattern as an early-warning layer for transport and utility planning.
- Review likely growth corridors against schools, healthcare, drainage and public-transport capacity.
- Identify non-built land that may face greater development pressure.
- Use the scenario alongside current plans and field evidence when prioritising detailed investigations.
- Update the scenario as newer imagery, demographic data and planning information become available.

Limitations

- Historical validation was 49.27% Overall Accuracy with Kappa 0.366.
- The 2035 map is scenario-based and not deterministic.
- Policy changes, infrastructure investment, land markets and demographic shifts may change future growth patterns.
- The output should not be used for parcel-level decisions without additional local evidence.

Tools

Google Earth Engine · GIS · Remote Sensing · CA-Markov · Suitability Modelling · Python · Spatial Analysis

Citation

Abdullah Abdazeez Ayomide (2026). Predicting Urban Growth to Support Better Planning - Abuja, Nigeria.